

Introduction to Artificial Intelligence

Knowledge

knowledge-based agents

agents that reason by operating on
internal representations of knowledge

If it didn't rain, Harry visited Hagrid today.

Harry visited Hagrid or Dumbledore today, but not both.

Harry visited Dumbledore today.

Harry did not visit Hagrid today.

It rained today.

Logic

sentence

an assertion about the world
in a knowledge representation language

Propositional Logic

Proposition Symbols

P

Q

R

Logical Connectives

$\neg$

not

$\wedge$

and

$\vee$

or

$\rightarrow$

implication

$\leftrightarrow$

biconditional

Not ($\neg$)

P	$\neg P$
false	true
true	false

And ($\wedge$)

P	Q	$P \wedge Q$
false	false	false
false	true	false
true	false	false
true	true	true

Or ($\vee$)

P	Q	$P \vee Q$
false	false	false
false	true	true
true	false	true
true	true	true

Implication ($\rightarrow$)

P	Q	$P \rightarrow Q$
false	false	true
false	true	true
true	false	false
true	true	true

Biconditional ($\leftrightarrow$)

P	Q	$P \leftrightarrow Q$
false	false	true
false	true	false
true	false	false
true	true	true

model

assignment of a truth value to every
propositional symbol (a "possible world")

model

P : It is raining.

Q : It is a Tuesday.

$\{P = \text{true}, Q = \text{false}\}$

knowledge base

a set of sentences known by a
knowledge-based agent

Entailment

$$\alpha \models \beta$$

In every model in which sentence α is true,
sentence β is also true.

If it didn't rain, Harry visited Hagrid today.

Harry visited Hagrid or Dumbledore today, but not both.

Harry visited Dumbledore today.

Harry did not visit Hagrid today.

It rained today.

inference

the process of deriving new sentences
from old ones

P: It is a Tuesday.

Q: It is raining.

R: Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ *P* $\neg Q$

Inference: *R*

Inference Algorithms

Does
 $KB \models \alpha$
?

Model Checking

Model Checking

- To determine if $KB \models \alpha$:
 - Enumerate all possible models.
 - If in every model where KB is true, α is true, then KB entails α .
 - Otherwise, KB does not entail α .

P : It is a Tuesday. Q : It is raining. R : Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ P $\neg Q$

Query: R

P	Q	R	KB
false	false	false	
false	false	true	
false	true	false	
false	true	true	
true	false	false	
true	false	true	
true	true	false	
true	true	true	

P : It is a Tuesday. Q : It is raining. R : Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ P $\neg Q$

Query: R

P	Q	R	KB
false	false	false	false
false	false	true	false
false	true	false	false
false	true	true	false
true	false	false	false
true	false	true	true
true	true	false	false
true	true	true	false

P : It is a Tuesday. Q : It is raining. R : Harry will go for a run.

KB: $(P \wedge \neg Q) \rightarrow R$ P $\neg Q$

Query: R

P	Q	R	KB
false	false	false	false
false	false	true	false
false	true	false	false
false	true	true	false
true	false	false	false
true	false	true	true
true	true	false	false
true	true	true	false

Knowledge Engineering

Clue



Clue

People

Col. Mustard

Prof. Plum

Ms. Scarlet

Rooms

Ballroom

Kitchen

Library

Weapons

Knife

Revolver

Wrench

Clue

People

Rooms

Weapons

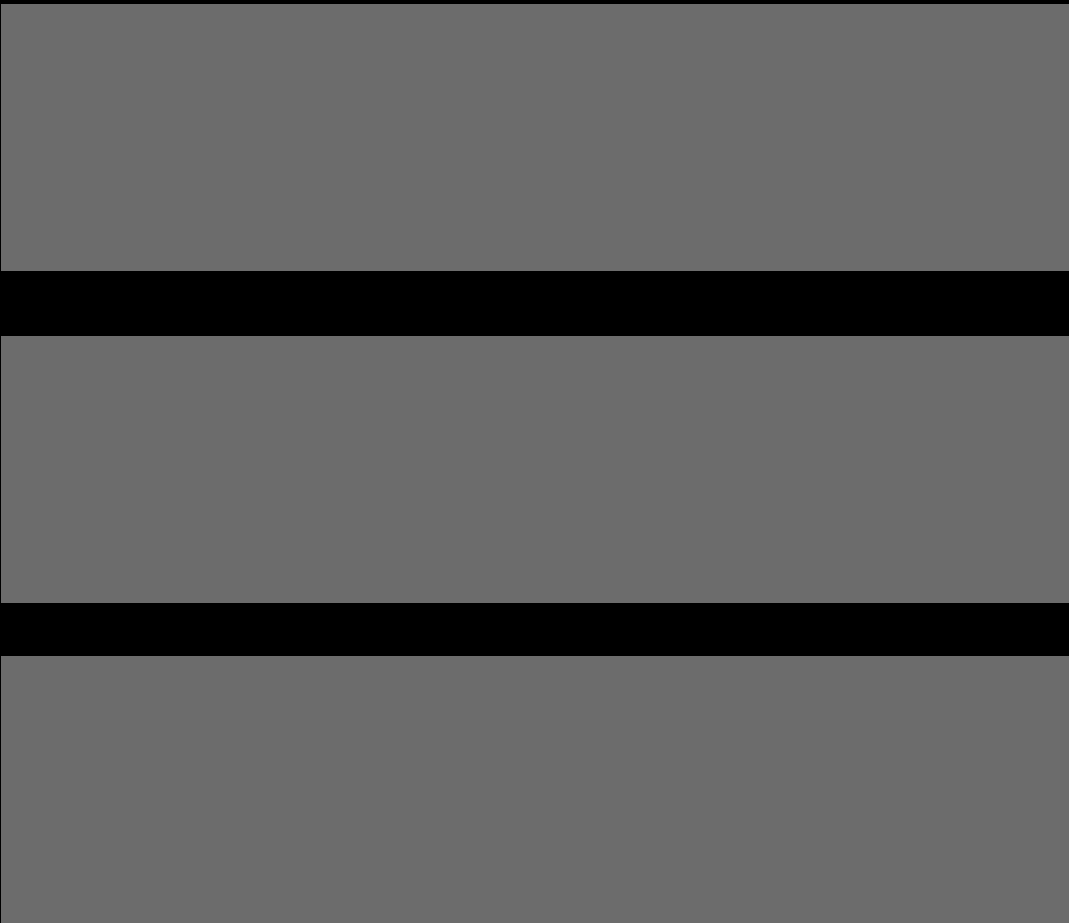
Clue

People

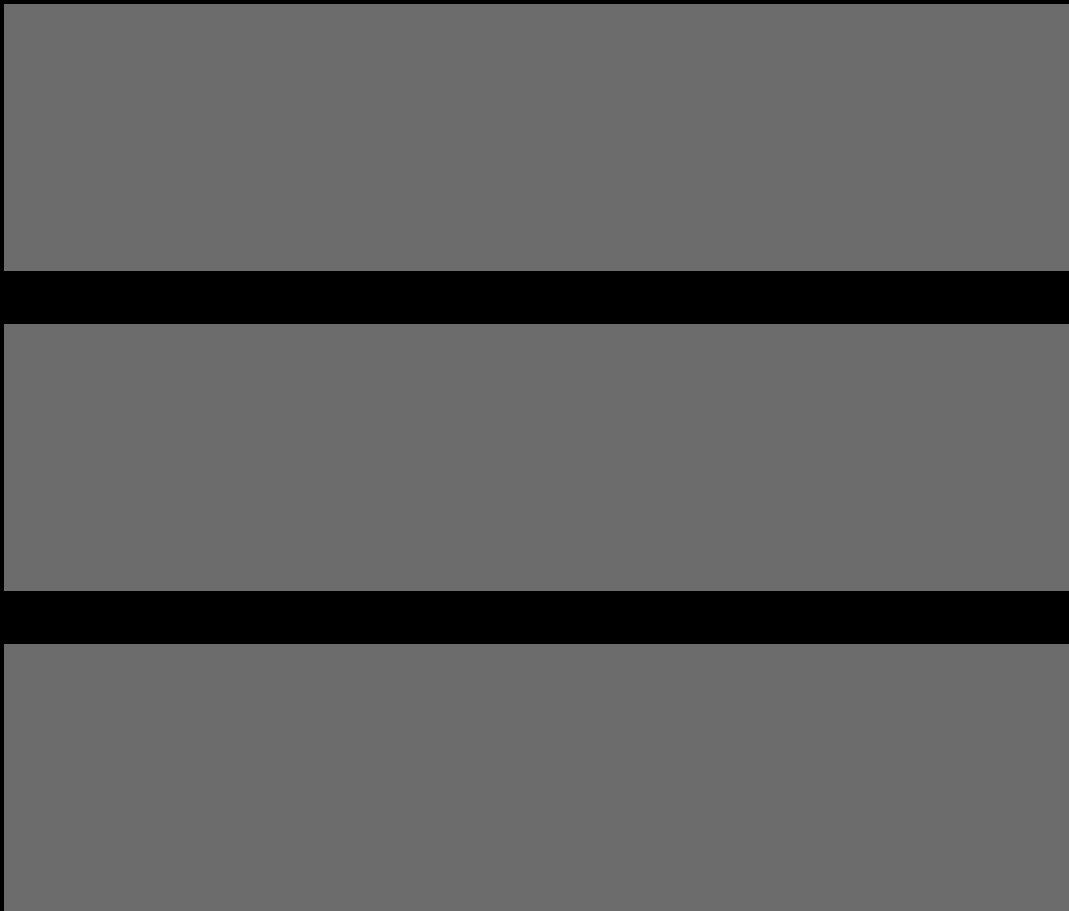
Rooms

Weapons

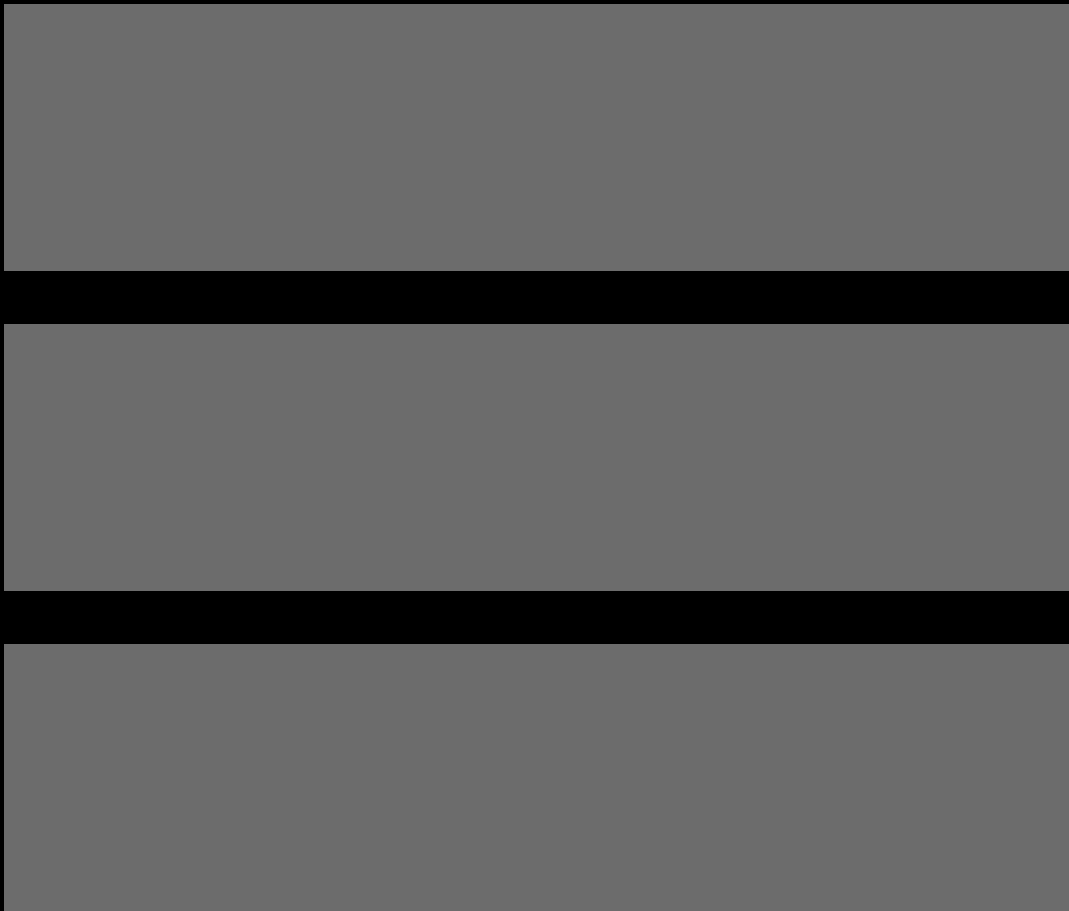
People



Rooms



Weapons



People



Rooms



Weapons



Clue

Propositional Symbols

mustard

ballroom

knife

plum

kitchen

revolver

scarlet

library

wrench

Clue

$(mustard \vee plum \vee scarlet)$

$(ballroom \vee kitchen \vee library)$

$(knife \vee revolver \vee wrench)$

$\neg plum$

$\neg mustard \vee \neg library \vee \neg revolver$

Logic Puzzles

- Gilderoy, Minerva, Pomona and Horace each belong to a different one of the four houses: Gryffindor, Hufflepuff, Ravenclaw, and Slytherin House.
- Gilderoy belongs to Gryffindor or Ravenclaw.
- Pomona does not belong in Slytherin.
- Minerva belongs to Gryffindor.

Logic Puzzles

Propositional Symbols

GilderoyGryffindor
GilderoyHufflepuff
GilderoyRavenclaw
GilderoySlytherin

PomonaGryffindor
PomonaHufflepuff
PomonaRavenclaw
PomonaSlytherin

MinervaGryffindor
MinervaHufflepuff
MinervaRavenclaw
MinervaSlytherin

HoraceGryffindor
HoraceHufflepuff
HoraceRavenclaw
HoraceSlytherin

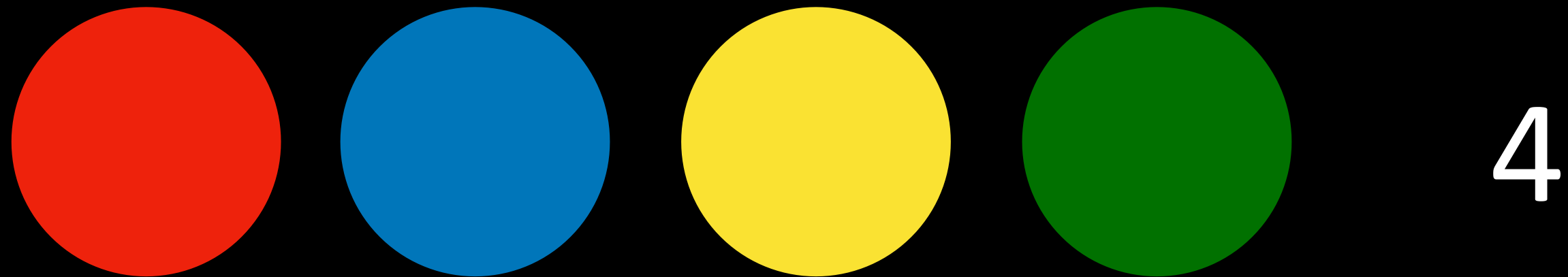
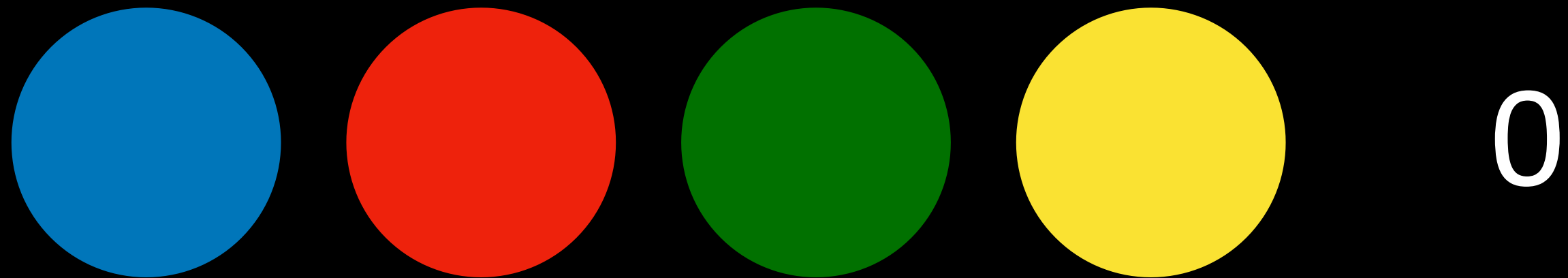
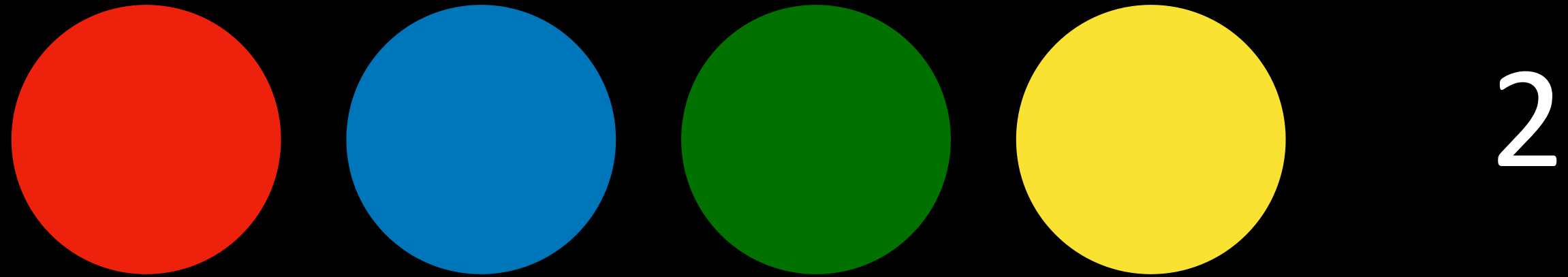
Logic Puzzles

(PomonaSlytherin $\rightarrow$ $\neg$ PomonaHufflepuff)

(MinervaRavenclaw $\rightarrow$ $\neg$ GilderoyRavenclaw)

(GilderoyGryffindor $\vee$ GilderoyRavenclaw)

Mastermind



Inference Rules

Modus Ponens

If it is raining, then Harry is inside.

It is raining.

Harry is inside.

Modus Ponens

$$\alpha \rightarrow \beta$$

$$\alpha$$



$$\beta$$

And Elimination

Harry is friends with Ron and Hermione.

Harry is friends with Hermione.

And Elimination

$$\alpha \wedge \beta$$

$$\alpha$$

Double Negation Elimination

It is not true that Harry did not pass the test.

Harry passed the test.

Double Negation Elimination

$$\neg(\neg\alpha)$$

$$\alpha$$

Implication Elimination

If it is raining, then Harry is inside.

It is not raining or Harry is inside.

Implication Elimination

$$\alpha \rightarrow \beta$$



$$\neg \alpha \vee \beta$$

Biconditional Elimination

It is raining if and only if Harry is inside.

If it is raining, then Harry is inside,
and if Harry is inside, then it is raining.

Biconditional Elimination

$$\alpha \leftrightarrow \beta$$

$$(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$$

De Morgan's Law

It is not true that both
Harry and Ron passed the test.

Harry did not pass the test
or Ron did not pass the test.

De Morgan's Law

$$\neg(\alpha \wedge \beta)$$

$$\neg\alpha \vee \neg\beta$$

De Morgan's Law

It is not true that
Harry or Ron passed the test.

Harry did not pass the test
and Ron did not pass the test.

De Morgan's Law

$$\neg(a \vee \beta)$$

$$\neg\alpha \wedge \neg\beta$$

Distributive Property

$$(a \wedge (\beta \vee \gamma))$$

$$(a \wedge \beta) \vee (a \wedge \gamma)$$

Distributive Property

$$(a \vee (\beta \wedge \gamma))$$

$$(a \vee \beta) \wedge (a \vee \gamma)$$

Search Problems

- initial state
- actions
- transition model
- goal test
- path cost function

Theorem Proving

- initial state: starting knowledge base
- actions: inference rules
- transition model: new knowledge base after inference
- goal test: check statement we're trying to prove
- path cost function: number of steps in proof

Resolution

$(\text{Ron is in the Great Hall}) \vee (\text{Hermione is in the library})$

Ron is not in the Great Hall

Hermione is in the library

$$P \vee Q$$

$$\neg P$$

$$Q$$

$$P \vee Q_1 \vee Q_2 \vee \dots \vee Q_n$$

$$\neg P$$

$$Q_1 \vee Q_2 \vee \dots \vee Q_n$$

$(\text{Ron is in the Great Hall}) \vee (\text{Hermione is in the library})$

$(\text{Ron is not in the Great Hall}) \vee (\text{Harry is sleeping})$

$(\text{Hermione is in the library}) \vee (\text{Harry is sleeping})$

$$P \vee Q$$

$$\neg P \vee R$$

$$Q \vee R$$

$$P \vee Q_1 \vee Q_2 \vee \dots \vee Q_n$$

$$\neg P \vee R_1 \vee R_2 \vee \dots \vee R_m$$

$$Q_1 \vee Q_2 \vee \dots \vee Q_n \vee R_1 \vee R_2 \vee \dots \vee R_m$$

clause

a disjunction of literals

e.g. $P \vee Q \vee R$

conjunctive normal form

logical sentence that is a conjunction of clauses

e.g. $(A \vee B \vee C) \wedge (D \vee \neg E) \wedge (F \vee G)$

Conversion to CNF

- Eliminate biconditionals
 - turn $(\alpha \leftrightarrow \beta)$ into $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$
- Eliminate implications
 - turn $(\alpha \rightarrow \beta)$ into $\neg\alpha \vee \beta$
- Move $\neg$ inwards using De Morgan's Laws
 - e.g. turn $\neg(\alpha \wedge \beta)$ into $\neg\alpha \vee \neg\beta$
- Use distributive law to distribute $\vee$ wherever possible

Conversion to CNF

$$(P \vee Q) \rightarrow R$$

$$\neg(P \vee Q) \vee R$$

eliminate implication

$$(\neg P \wedge \neg Q) \vee R$$

De Morgan's Law

$$(\neg P \vee R) \wedge (\neg Q \vee R)$$

distributive law

Inference by Resolution

$$P \vee Q$$

$$\neg P \vee R$$

$$(Q \vee R)$$

$$P \vee Q \vee S$$

$$\neg P \vee R \vee S$$

$$(Q \vee S \vee R \vee S)$$

$$P \vee Q \vee S$$

$$\neg P \vee R \vee S$$

$$(Q \vee R \vee S)$$

P

$\neg P$



$()$

Inference by Resolution

- To determine if $KB \models \alpha$:
 - Check if $(KB \wedge \neg\alpha)$ is a contradiction?
 - If so, then $KB \models \alpha$.
 - Otherwise, no entailment.

Inference by Resolution

- To determine if $KB \models \alpha$:
 - Convert $(KB \wedge \neg\alpha)$ to Conjunctive Normal Form.
 - Keep checking to see if we can use resolution to produce a new clause.
 - If ever we produce the **empty** clause (equivalent to False), we have a contradiction, and $KB \models \alpha$.
 - Otherwise, if we can't add new clauses, no entailment.

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A)$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad \underline{(\neg B \vee C)} \quad \underline{(\neg C)} \quad (\neg A)$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad \underline{(\neg B \vee C)} \quad \underline{(\neg C)} \quad (\neg A) \quad (\neg B)$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad (\neg B)$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$\underline{(A \vee B)} \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad \underline{(\neg B)}$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$\underline{(A \vee B)} \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad \underline{(\neg B)} \quad (A)$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad (\neg B) \quad (A)$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad \underline{(\neg A)} \quad (\neg B) \quad \underline{(A)}$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad \underline{(\neg A)} \quad (\neg B) \quad \underline{(A)} \quad ()$$

Inference by Resolution

Does $(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C)$ entail A ?

$$(A \vee B) \wedge (\neg B \vee C) \wedge (\neg C) \wedge (\neg A)$$

$$(A \vee B) \quad (\neg B \vee C) \quad (\neg C) \quad (\neg A) \quad (\neg B) \quad (A) \quad ()$$

First-Order Logic

Propositional Logic

Propositional Symbols

MinervaGryffindor

MinervaHufflepuff

MinervaRavenclaw

MinervaSlytherin

...

First-Order Logic

Constant Symbol

Minerva

Pomona

Horace

Gilderoy

Gryffindor

Hufflepuff

Ravenclaw

Slytherin

Predicate Symbol

Person

House

BelongsTo

First-Order Logic

Person(Minerva)

Minerva is a person.

House(Gryffindor)

Gryffindor is a house.

$\neg$ *House(Minerva)*

Minerva is not a house.

BelongsTo(Minerva, Gryffindor)

Minerva belongs to Gryffindor.

Universal Quantification

Universal Quantification

$$\forall x. \textit{BelongsTo}(x, \textit{Gryffindor}) \rightarrow \\ \neg \textit{BelongsTo}(x, \textit{Hufflepuff})$$

For all objects x , if x belongs to Gryffindor,
then x does not belong to Hufflepuff.

Anyone in Gryffindor is not in Hufflepuff.

Existential Quantification

Existential Quantification

$$\exists x. \textit{House}(x) \wedge \textit{BelongsTo}(\textit{Minerva}, x)$$

There exists an object x such that
 x is a house and Minerva belongs to x .

Minerva belongs to a house.

Existential Quantification

$$\forall x. \textit{Person}(x) \rightarrow (\exists y. \textit{House}(y) \wedge \textit{BelongsTo}(x, y))$$

For all objects x , if x is a person, then there exists an object y such that y is a house and x belongs to y .

Every person belongs to a house.

Knowledge

Introduction to Artificial Intelligence